

1. Introduction to Dipoles and Surface Charge Distributions.

In electrostatics it is very common to encounter problems where the electric field is discontinuous. The most common scenerio where this happens is when we have a charge distribution over a surface. Let us consider a surface S and let $\vec{E}_1(\vec{r})$ is the electric field entering through one side of the surface at position $\vec{r}$ and $\vec{E}_2(\vec{r})$ the electric field leaving the surface at the other side. If we focus on a small area around position $\vec{r}$ we can use Gauss's Law and write

$$(\vec{E}_1(\vec{r}) - \vec{E}_2(\vec{r})) \cdot \hat{n} = \frac{\sigma}{\epsilon_0}$$

. This shows that a simple charged surface can introduce a discontinuity of $\frac{\sigma}{\epsilon_0}$ in the direction normal to the surface at any position in the charged surface. The potential or electric field generated due to this small area on the surface S at position $\vec{r}$ can be easily calculated. The potential at any position $\vec{r}'$ due this charged surface S can then be written as

$$\Phi(\vec{r}') = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r})}{\|\vec{r}' - \vec{r}\|} d^2\vec{r}$$

We also consider the scenerio below in Figure 1. We have two charged surfaces, in our case oppositely charged, very close to each other. This is called a dipole layer charged distribution. The strength of the dipole-layer distribution is represented by $D(\vec{r})$ which is given by

$$D(\vec{r}) = \lim_{d\vec{r} \rightarrow 0} \sigma(\vec{r}) d\vec{r}$$

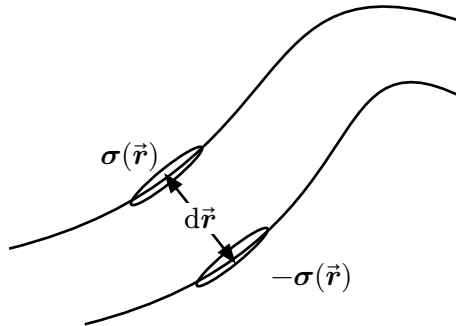


Figure 1: The figure shows two surfaces which are $d\vec{r}$ apart at position $\vec{r}$ and both surfaces have opposite charges but the magnitude of the charge density is given by the function σ .

Now we already have calculated the potential at an arbitrary position $\vec{r}'$ due to a surface with a charge distributed on it. We can use the same concept to calculate the potential at any arbitrary position $\vec{r}'$ due to a dipole-layer. The potential at position $\vec{r}'$ can be written as

$$\Phi(\vec{r}') = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r})}{\|\vec{r}' - \vec{r}\|} d^2\vec{r} - \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r})}{\|\vec{r}' - (\vec{r} + d\vec{r})\|} d^2\vec{r}$$

Now, for two surfaces that are very close to each other and with the dipole moment denoted by $\vec{p}$ we can write

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{r}' - \vec{r})}{\|\vec{r}' - \vec{r}\|^3}$$